MATH 6660 — Homework 2

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Each step is tagged with the lecture it comes from. Anything outside the lectures is boxed and labelled **NOT FROM LECTURE**.

Problem 1

Problem. For

$$\textbf{ODE}: \ -u'' + q(x)\,u = f(x), \qquad \textbf{BC1}: \ u(-\pi) = u(\pi), \qquad \textbf{BC2}: \ u'(-\pi) = u'(\pi), \tag{1}$$

with $q > 0$ on $[-\pi, \pi]$: derive a Galerkin orthogonality expression, and deduce from it an appropriate energy norm.

Step 1 — variational form [Lecture 1]

- Multiply (1) by a test function v and integrate:

$$\int_{-\pi}^{\pi} \big(-u'' + q\,u\big) v \,\mathrm{d}x = \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x. \tag{2}$$

- Integrate the u'' term by parts once:

$$\int_{-\pi}^{\pi} \big(u'v' + q\,u\,v\big)\,\mathrm{d}x \ - \ \big[\,u'v\,\big]_{-\pi}^{\pi} \ = \ \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x. \tag{3}$$

- Collapse the boundary bracket with **BC2**, $u'(-\pi) = u'(\pi)$:

$$\big[\,u'v\,\big]_{-\pi}^{\pi} = u'(\pi)\,v(\pi) - u'(-\pi)\,v(-\pi) = u'(\pi)\big[\,v(\pi) - v(-\pi)\,\big]. \tag{4}$$

- It vanishes exactly when v is periodic. That requirement defines the space:

$$V = \Big\{\, v : \ \int_{-\pi}^{\pi} \big(v^2 + (v')^2\big)\,\mathrm{d}x < \infty, \ \ v(-\pi) = v(\pi) \,\Big\}. \tag{5}$$

- **Variational problem:** find $u \in V$ such that

$$\int_{-\pi}^{\pi} \big(u'v' + q\,u\,v\big)\,\mathrm{d}x = \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x \qquad \forall\, v \in V. \tag{6}$$

- **BC1** essential — built into V in (5). **BC2** natural — not imposed on V; it is what collapsed (4). [Lecture 6, ① ②]
- Trial space = test space = V.

Step 2 — the finite element method [Lecture 6]

- Take any $V_h \subset V$. Find $U \in V_h$ such that

$$\int_{-\pi}^{\pi} \big(U'v' + q\,U\,v\big)\,\mathrm{d}x = \int_{-\pi}^{\pi} f\,v\,\mathrm{d}x \qquad \forall\, v \in V_h. \tag{7}$$

Step 3 — Galerkin orthogonality [Lecture 6]

- $V_h \subset V$, so (6) holds in particular for every $v \in V_h$.
- Subtract (7) from it — the right-hand sides cancel.
- With $e := u - U$:

$$\boxed{\int_{-\pi}^{\pi} \big(e'v' + q\,e\,v\big)\,\mathrm{d}x = 0 \qquad \forall\, v \in V_h} \tag{8}$$

- The v in (8) is a **dummy slot**. The statement is that this integral vanishes whenever the slot holds a member of V_h — whatever we happen to call it. Later steps fill that slot with specific members.

Step 4 — energy norm, read off (8) [Lecture 6]

- The integrand of (8) pairs *two* functions, e and v:

$$e'v' + q\,e\,v.$$

- The energy norm is that same integrand with **both factors the same function**. Put v in place of both:

$$\int_{-\pi}^{\pi} \big(v'v' + q\,v\,v\big)\,\mathrm{d}x = \int_{-\pi}^{\pi} \Big[\,(v')^2 + q\,v^2\,\Big]\,\mathrm{d}x \;\geq\; 0.$$

  [Lecture 6 does exactly this for $-(au')' = f$: Galerkin orthogonality there is $\int_0^1 a\,(u-U)'v'\,\mathrm{d}x = 0$, and the energy norm is $\|v\|_E^2 = \int_0^1 a\,v'v'\,\mathrm{d}x$ — the same integrand with both factors v]

- Hence

$$\boxed{\|v\|_E := \left(\int_{-\pi}^{\pi} \Big[\,(v')^2 + q\,v^2\,\Big]\,\mathrm{d}x\right)^{1/2}} \tag{9}$$

- Why this is the *appropriate* one: it was not chosen. It is the integrand of (8) itself, which came out of the integration by parts in Step 1. Change the ODE and the integrand changes, so the norm changes with it.
- It is a norm: both terms of the integrand are non-negative, so $\|v\|_E = 0$ forces $\int (v')^2\,\mathrm{d}x = 0$ *and* $\int q\,v^2\,\mathrm{d}x = 0$; with $q > 0$ this gives $v \equiv 0$. [Lecture 3 runs the same argument with $a \geq a_1 > 0$]
- Why the problem states $q > 0$: with $q \equiv 0$ this would be only a *seminorm*. Periodic conditions leave constants in (5), and a constant has $v' \equiv 0$ but $v \not\equiv 0$. The $q\,v^2$ term is what removes them.

Problem 2

Problem. Develop a CG(1) method for (1), clearly stating the appropriate spaces, trial, test, and basis functions.

Everything below is the Lecture 3–4 construction applied to (1). Two things differ from the lecture, and both are forced by this problem rather than imported from anywhere: the bilinear expression carries the extra $q\,uv$ term (from Problem 1, Step 1), and the boundary conditions are periodic instead of Dirichlet.

Step 1 — mesh [Lecture 3: defines a mesh $\mathcal{T}_h$ on the interval with $h_j = x_j - x_{j-1}$, spacing allowed to vary]

- Take the uniform mesh of the hint:

$$h = \frac{2\pi}{M+1}, \qquad x_j = -\pi + j\,h, \quad j = 0, 1, \ldots, M+1. \tag{10}$$

- So $x_0 = -\pi$ and $x_{M+1} = \pi$. Elements $I_j = [x_{j-1}, x_j]$, $j = 1, \ldots, M+1$ — that is $M+1$ elements.
- **BC1 says x_0 and x_{M+1} are the same point of the domain.** So there are $M+1$ *distinct* nodes, not $M+2$.

Step 2 — the space V_h [Lecture 3: "the simplest $V_h \subset V$ is the space of continuous piecewise linear polynomials on $\mathcal{T}_h$" — this is CG(1)]

- Continuous piecewise linears on this mesh:

$$V_h = \left\{ v \in V : \; v|_{I_j} \text{ is linear}, \; j = 1, \ldots, M+1 \right\}. \tag{11}$$

- $V_h \subset V$ by construction. That containment is what Step 3 of Problem 1 needed in order to subtract, so Galerkin orthogonality (8) applies here unchanged.
- $\dim V_h = M+1$: a continuous piecewise linear on $M+1$ elements carries $M+2$ nodal values, and BC1 identifies the two end values as one. [Lecture 4 gets $\dim V_h = M$ for Dirichlet: there the two end values are *deleted*; here they are *merged*]

Step 3 — basis functions [Lecture 3–4: the hat function φ_j associated with node x_j, and $V_h = \operatorname{span}\{\varphi_1, \ldots, \varphi_M\}$]

- Interior nodes $j = 1, \ldots, M$ — the lecture's hat, with $h_j \equiv h$:

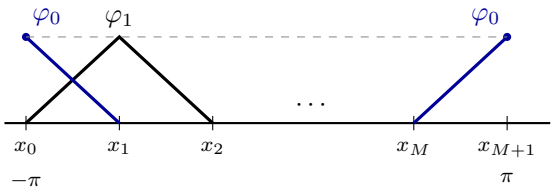
$$\varphi_j(x) = \begin{cases} \dfrac{x - x_{j-1}}{h}, & x_{j-1} \le x \le x_j, \\ \dfrac{x_{j+1} - x}{h}, & x_j \le x \le x_{j+1}, \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

- At the glued node, one hat whose two halves sit at the two ends of the interval:

$$\varphi_0(x) = \begin{cases} \dfrac{x_1 - x}{h}, & x_0 \le x \le x_1, \\ \dfrac{x - x_M}{h}, & x_M \le x \le x_{M+1}, \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

- Check: $\varphi_0(x_0) = 1$ and $\varphi_0(x_{M+1}) = 1$ — the same value at the same point of the domain, so φ_0 is continuous there and $\varphi_0 \in V_h$. Nodal property $\varphi_i(x_j) = \delta_{ij}$ holds for all $i, j = 0, \ldots, M$.
- Hence the basis, one hat per distinct node:

$$V_h = \operatorname{span}\{\varphi_0, \varphi_1, \ldots, \varphi_M\}, \qquad \dim V_h = M + 1. \tag{14}$$



The two blue halves are *one* basis function φ_0, carrying the single unknown ξ_0: x_0 and x_{M+1} are the same point of the domain. φ_1 is an ordinary interior hat.

Step 4 — trial and test functions [Lecture 3: any $U \in V_h$ is uniquely $U = \sum_j \xi_j \varphi_j$; Lecture 4: take $v = \varphi_i$]

- **Trial:**

$$U(x) = \sum_{j=0}^{M} \xi_j \, \varphi_j(x), \qquad \xi_j = U(x_j). \tag{15}$$

- **Test:** $v = \varphi_i, \;\; i = 0, 1, \ldots, M.$
- Trial space = test space = V_h — this is what makes it Galerkin.

Step 5 — the linear system [Lecture 4: substitute the expansion into the FEM statement, take $v = \varphi_i$, obtain $\boldsymbol{K\xi} = \boldsymbol{b}$ with $K_{ij} = \int a\,\varphi_i'\varphi_j'\,\mathrm{d}x$, $b_i = \int f\varphi_i\,\mathrm{d}x$]

- Put (15) into the FEM statement (7) and take $v = \varphi_i$:

$$\sum_{j=0}^{M} \xi_j \int_{-\pi}^{\pi} \left(\varphi_i'\varphi_j' + q\,\varphi_i\varphi_j\right) \mathrm{d}x = \int_{-\pi}^{\pi} f\,\varphi_i\,\mathrm{d}x, \qquad i = 0, 1, \ldots, M.$$

- That is an $(M+1) \times (M+1)$ system:

$$\boxed{\boldsymbol{K\xi} = \boldsymbol{b}, \qquad K_{ij} = \int_{-\pi}^{\pi} \left(\varphi_i'\varphi_j' + q\,\varphi_i\varphi_j\right) \mathrm{d}x, \qquad b_i = \int_{-\pi}^{\pi} f\,\varphi_i\,\mathrm{d}x} \tag{16}$$

- **Symmetric**, because test and trial come from the same space. [Lecture 4 lists this as property (1) of $\boldsymbol{K}$]
- **Sparse**: φ_i is nonzero on two elements only, so $K_{ij} = 0$ unless φ_i and φ_j share an element. [Lecture 4, property (2)]
- **Positive definite**: $\sum_{i,j} \eta_i K_{ij} \eta_j = \|w\|_E^2 > 0$ for $w = \sum_i \eta_i \varphi_i \not\equiv 0$, by (9). So $\boldsymbol{K}$ is invertible and U exists and is unique. [Lecture 4, property (3)]
- **Periodic bookkeeping:** φ_0 and φ_M both live on the last element $[x_M, x_{M+1}]$, so $K_{0,M} = K_{M,0}$ is an entry of the matrix. The pattern is tridiagonal *plus two corners* — no row or column is deleted, unlike the Dirichlet case of Lecture 4.

Problem 3

Problem. Show that the CG(1) solution is optimal in the energy norm (9), and use this to prove an *a priori* error estimate in the energy norm.

Notation. $e := u - U$. Two norms appear, told apart by the subscript:

$$\|v\|_E = \Big(\int_{-\pi}^{\pi} \big[(v')^2 + q\,v^2\big]\,\mathrm{d}x\Big)^{1/2} \quad \text{(energy, eq. 9)}, \qquad \|v\| = \Big(\int_{-\pi}^{\pi} v^2\,\mathrm{d}x\Big)^{1/2} \quad (L^2).$$

Lecture 6–7 write $\|\cdot\|_a$ for the a-weighted L^2 norm; here the coefficient in front of u'' is 1, so $\|\cdot\|_a = \|\cdot\|$. Standing assumption: q is bounded, $0 < q \le q_{\max} < \infty$ — true for the data of Problem 5.

Step 1 — Cauchy's inequality [Lecture 6]

- Standard form, as stated in Lecture 6:

$$\left|\int_0^1 a\,v'w'\,\mathrm{d}x\right| \;\le\; \|v\|_E\,\|w\|_E.$$

- For this problem ($a = 1$, with the extra $q\,v\,w$ term):

$$\boxed{\left|\int_{-\pi}^{\pi} \big(v'w' + q\,v\,w\big)\,\mathrm{d}x\right| \;\le\; \|v\|_E\,\|w\|_E} \tag{17}$$

- where, written out from (9),

$$\|v\|_E = \left(\int_{-\pi}^{\pi}\Big[\,(v')^2 + q\,v^2\,\Big]\,\mathrm{d}x\right)^{1/2}, \qquad \|w\|_E = \left(\int_{-\pi}^{\pi}\Big[\,(w')^2 + q\,w^2\,\Big]\,\mathrm{d}x\right)^{1/2}.$$

Step 2 — optimality [Lecture 6: for any $v \in V_h$ insert $-v + v$, kill the second term by Galerkin orthogonality, apply Cauchy, cancel one factor — the boxed result there is $\|u - U\|_E \le \|u - v\|_E$]

- Start from the definition of the energy norm, both factors $u - U$:

$$\|u - U\|_E^2 = \int_{-\pi}^{\pi}\Big[(u-U)'(u-U)' + q\,(u-U)(u-U)\Big]\,\mathrm{d}x.$$

- For *any* $v \in V_h$, insert $-v + v$ in the second factor:

$$\begin{aligned}
\|u - U\|_E^2 &= \int_{-\pi}^{\pi}\Big[(u-U)'\,(u - v + v - U)' + q\,(u-U)(u - v + v - U)\Big]\,\mathrm{d}x \\
&= \int_{-\pi}^{\pi}\Big[(u-U)'(u-v)' + q\,(u-U)(u-v)\Big]\,\mathrm{d}x \\
&\quad + \underbrace{\int_{-\pi}^{\pi}\Big[(u-U)'(v-U)' + q\,(u-U)(v-U)\Big]\,\mathrm{d}x}_{=\,0 \text{ by Galerkin orthogonality (8)}}.
\end{aligned} \tag{18}$$

- **Why the second integral vanishes.** Compare it with (8):

$$\int_{-\pi}^{\pi}\big(e'\,\square' + q\,e\,\square\big)\,\mathrm{d}x = 0 \quad \text{for any } \square \in V_h, \qquad \text{here } \square = v - U.$$

  And $v - U \in V_h$, because v and U both lie in V_h and V_h is a vector space. So (8) applies and the integral is 0.

- By contrast $u - v \notin V_h$ (u is not piecewise linear), so (8) says nothing about the first integral. That is exactly why the split is arranged to put the V_h member in the *second* factor. [Lecture 6 makes exactly this point]
- So, for any $v \in V_h$,

$$\|u-U\|_E^2 = \int_{-\pi}^{\pi} \Big[(u-U)'(u-v)' + q\,(u-U)(u-v)\Big]\,\mathrm{d}x \;\le\; \|u-U\|_E\,\|u-v\|_E$$

  by (17).
- Cancel one factor $\|u-U\|_E$ (if it is zero there is nothing to prove):

$$\boxed{\|u-U\|_E \;\le\; \|u-v\|_E \qquad \forall\, v \in V_h} \tag{19}$$

- In words: U is the *best* member of V_h in the energy norm — no other function the mesh can hold is closer to u.

Step 3 — choose the interpolant

[Lecture 7: "take the test function $v = \Pi_h u$", the piecewise linear that agrees with u at every node — (19) holds for every v, so it holds for the one we can estimate]

- Take the nodal interpolant — the piecewise linear through u's nodal values:

$$\Pi_h u = \sum_{j=0}^{M} u(x_j)\,\varphi_j \in V_h.$$

- It is genuinely in V_h: continuous piecewise linear by construction, and the single unknown at the glued node is assigned $u(x_0) = u(x_{M+1})$ — consistent, because BC1 says those are the same value.
- So (19) with $v = \Pi_h u$ gives $\|u-U\|_E \le \|u-\Pi_h u\|_E$, and the solver has left the problem entirely. What remains is a question about *approximation*.

Step 4 — bound the interpolation error

Write $w := u - \Pi_h u$, so $\|w\|_E^2 = \int (w')^2\,\mathrm{d}x + \int q\,w^2\,\mathrm{d}x$. Bound the two terms separately.

- **Derivative term.** [Lecture 7: $\|u-\Pi_h u\|_E = \|(u-\Pi_h u)'\|_a \le C\|h\,u''\|_a$, "which just comes from approximating u' by a piecewise constant"]

$$\|w'\| \;\le\; C\,\|h\,u''\| \;=\; C\,h\,\|u''\| \qquad (h \text{ constant here}). \tag{20}$$

- **Function term.** w vanishes at both ends of every element, since $\Pi_h u$ matches u at the nodes. On element I_j, for $x \in I_j$,

$$w(x) = w(x_{j-1}) + \int_{x_{j-1}}^{x} w'\,\mathrm{d}x = \int_{x_{j-1}}^{x} w'\,\mathrm{d}x,$$

  so by Cauchy's inequality [Lecture 6's second Cauchy inequality, $\left|\int vw\,\mathrm{d}x\right| \le \|v\|_a\|w\|_{a^{-1}}$; with $a=1$ this is $\left|\int vw\,\mathrm{d}x\right| \le \|v\|\|w\|$, applied here to $\int 1\cdot w'$]

$$|w(x)| \le \int_{I_j} |w'|\,\mathrm{d}x \le \Big(\int_{I_j} 1\,\mathrm{d}x\Big)^{1/2}\Big(\int_{I_j}(w')^2\,\mathrm{d}x\Big)^{1/2} = \sqrt{h}\,\|w'\|_{I_j}.$$

  Hence $\int_{I_j} w^2\,\mathrm{d}x \le h\cdot\max_{I_j} w^2 \le h^2\|w'\|_{I_j}^2$, and summing over the elements

$$\|w\| \;\le\; h\,\|w'\|. \tag{21}$$

- **Combine.**

$$\begin{aligned}
\|w\|_E^2 = \|w'\|^2 + \int q\, w^2 \,\mathrm{d}x \;&\le\; \|w'\|^2 + q_{\max}\|w\|^2 && q \le q_{\max} \\
&\le\; \big(1 + q_{\max} h^2\big)\|w'\|^2 && \text{by (21)} \\
&\le\; \big(1 + q_{\max} h^2\big)\, C^2 h^2\, \|u''\|^2 && \text{by (20)}
\end{aligned}$$

Result

Chaining (19) with $v = \Pi_h u$ and Step 4, and using $h = 2\pi/(M+1) \le 2\pi$ to clear the remaining h from the constant:

$$\boxed{\|u - U\|_E \;\le\; \|u - \Pi_h u\|_E \;\le\; \widetilde{C}\, h\, \|u''\|, \qquad \widetilde{C} = C\sqrt{1 + 4\pi^2 q_{\max}}} \tag{22}$$

- So the CG(1) energy-norm error is $O(h)$: **first order**. [Lecture 7 states the same conclusion for its problem: "CG(1) is first order accurate in the energy norm, provided $u'' \in L^2$"]
- $\widetilde{C}$ depends only on $q_{\max}$ and the interpolation constant — not on the mesh.
- Two separate facts did the work. (19) is about the *method*: Galerkin always picks the best member of V_h. (20) is about the *space*: how well piecewise linears can represent u at all. The method contributes no error of its own.

Problem 4

Problem. Prove an *a priori* error estimate in the L^2-norm.
$e := u - U$, $\|v\| = \big(\int_{-\pi}^{\pi} v^2 \,\mathrm{d}x\big)^{1/2}$. Standing assumption, now needed at both ends: $0 < q_{\min} \le q \le q_{\max} < \infty$ (true for the data of Problem 5, where $q_{\min} = 2 - \sqrt{2}$ and $q_{\max} = 4.25$). The energy estimate (22) is only $O(h)$; the point of this problem is to gain one more power of h.

Step 1 — the dual problem [Lecture 7: "consider the following *dual problem*" $-(a\phi')' = e$, $\phi(0) = \phi(1) = 0$, where $e = u - U$; and the note that because the BVP is self-adjoint, the primal and dual operators are the same]

- Same construction, our operator and our boundary conditions — let ϕ solve

$$-\phi'' + q(x)\,\phi = e, \qquad \phi(-\pi) = \phi(\pi), \qquad \phi'(-\pi) = \phi'(\pi). \tag{23}$$

- The load of the dual problem is the error itself. That is the whole trick: pairing ϕ against e will reproduce $\|e\|^2$.

Step 2 — turn $\|e\|^2$ into an integral against ϕ [Lecture 7: multiply the dual problem by e and integrate; the boundary term dies and leaves $\|e\|^2 = \int a\, e'\phi' \,\mathrm{d}x$]

- Multiply (23) by e and integrate:

$$\|e\|^2 = \int_{-\pi}^{\pi} e^2 \,\mathrm{d}x = \int_{-\pi}^{\pi} \big(-\phi'' + q\,\phi\big) e \,\mathrm{d}x.$$

- Integrate the ϕ'' term by parts once:

$$\|e\|^2 = \int_{-\pi}^{\pi} \big(\phi' e' + q\,\phi\, e\big) \,\mathrm{d}x - \big[\,\phi' e\,\big]_{-\pi}^{\pi}.$$

- The boundary term vanishes: by the dual's own BC2, $\phi'(-\pi) = \phi'(\pi)$, so it collapses to $\phi'(\pi)\big[e(\pi) - e(-\pi)\big]$, and $e = u - U$ is periodic because u and U both are. Exactly the collapse of Problem 1, Step 1.

$$\|e\|^2 = \int_{-\pi}^{\pi} \big(e'\phi' + q\, e\, \phi\big) \,\mathrm{d}x. \tag{24}$$

Step 3 — insert Galerkin orthogonality [Lecture 7: "Recall Galerkin orthogonality ... Take $v = \Pi_h\phi$", which subtracts nothing but converts ϕ' into the interpolation error $(\phi - \Pi_h\phi)'$ — "that single substitution is what buys the extra power of h"]

- Fill the dummy slot of (8) with $\Pi_h\phi \in V_h$:

$$\int_{-\pi}^{\pi} \Big(e'(\Pi_h\phi)' + q\, e\, (\Pi_h\phi)\Big) \,\mathrm{d}x = 0.$$

- Subtract it from (24) — it takes nothing away, but it replaces ϕ by $\phi - \Pi_h\phi$:

$$\|e\|^2 = \int_{-\pi}^{\pi} \Big(e'\,(\phi - \Pi_h\phi)' + q\, e\, (\phi - \Pi_h\phi)\Big) \,\mathrm{d}x. \tag{25}$$

- ϕ' is not small; $(\phi - \Pi_h\phi)'$ is $O(h)$. That is where the extra power of h comes from.

Step 4 — Cauchy, then interpolation [Lecture 7: apply Cauchy, then the interpolation estimate, then recall the energy bound from part (1)]

- Cauchy (17) on (25):
$$\|e\|^2 \;\le\; \|e\|_E \, \|\phi - \Pi_h \phi\|_E .$$
- Step 4 of Problem 3 bounded $\|u - \Pi_h u\|_E$; nothing there used any property of u beyond having a second derivative, so the same bound holds for ϕ:
$$\|\phi - \Pi_h \phi\|_E \;\le\; \widetilde{C}\, h \, \|\phi''\| .$$
- And $\|e\|_E \le \widetilde{C}\, h \, \|u''\|$ is (22). So
$$\|e\|^2 \;\le\; \widetilde{C}^{\,2} h^2 \, \|u''\| \, \|\phi''\| . \tag{26}$$
- Only $\|\phi''\|$ still refers to the unknown ϕ. Trade it for $\|e\|$ and the estimate closes.

Step 5 — the stability constant [Lecture 8: bound $\|\psi''\|$ by reading ψ'' off the equation, $-a\psi'' = a'\psi' + \xi$, then bounding the lower-order term by multiplying the BVP by ψ and integrating by parts]

- **Read ϕ'' off (23):** $\phi'' = q\,\phi - e$, so
$$\|\phi''\| \;\le\; q_{\max}\|\phi\| + \|e\| .$$
- **Bound $\|\phi\|$.** Multiply (23) by ϕ and integrate by parts, exactly as in Step 2 (the boundary term dies the same way):
$$\int_{-\pi}^{\pi} \Big[(\phi')^2 + q\,\phi^2 \Big] \, \mathrm{d}x = \int_{-\pi}^{\pi} e\,\phi \, \mathrm{d}x \;\le\; \|e\| \, \|\phi\|$$
by Cauchy. The left side is $\|\phi\|_E^2$, and since $(\phi')^2 \ge 0$ and $q \ge q_{\min}$,
$$q_{\min}\|\phi\|^2 \;\le\; \|\phi\|_E^2 \;\le\; \|e\| \, \|\phi\| ,$$
$$\|\phi\| \;\le\; \frac{\|e\|}{q_{\min}} . \tag{27}$$
- **Combine:**
$$\|\phi''\| \;\le\; q_{\max}\frac{\|e\|}{q_{\min}} + \|e\| = \underbrace{\Big(1 + \frac{q_{\max}}{q_{\min}}\Big)}_{=:\,C_S} \|e\| . \tag{28}$$
- [Lecture 8 needs the Poincaré inequality at this point, because its problem has no zeroth-order term and $\|\psi\|$ must be got from $\|\psi'\|$. Here the $q\,\phi^2$ term bounds $\|\phi\|$ directly, so no Poincaré is needed — but $q_{\min} > 0$ is, which is why it was assumed]

Result

Put (28) into (26):
$$\|e\|^2 \;\le\; \widetilde{C}^{\,2} C_S \, h^2 \, \|u''\| \, \|e\| ,$$
and cancel one factor $\|e\|$ (if it is zero there is nothing to prove):

$$\boxed{\|u - U\| \;\le\; \widetilde{C}^{\,2} C_S \, h^2 \, \|u''\| , \qquad C_S = 1 + \frac{q_{\max}}{q_{\min}}} \tag{29}$$

- So the CG(1) error in L^2 is $O(h^2)$: **second order** — one power better than the energy norm (22).
- Where the extra power came from: (25) pairs the error against the *interpolation error of* ϕ rather than against ϕ itself, and interpolation costs one power of h.